

CS League Platform — Competitive CS Tournament System

React

FastAPI

PostgreSQL

Vercel

Render

License

Современная fullstack платформа для проведения CS-турниров с системой ELO-рейтинга, placement-калибровкой, live veto системой, турнирными сетками, Hall of Fame и полноценной backend-архитектурой.

Live Demo

Frontend

<https://cs-league-16-55.vercel.app/>

Backend API

<https://cs-league-16-55.onrender.com/>

Swagger Documentation

<https://cs-league-16-55.onrender.com/docs>

GitHub Repository

https://github.com/TursunovJavoxir/CS_League_16-55

Основные Возможности

Authentication & Security

- JWT Authentication
 - Регистрация и авторизация игроков
 - Admin Role System
 - Protected Routes
 - Password Hashing (bcrypt)
 - Token-based authorization
 - Защита admin endpoints
 - CORS protection
-

Competitive System

ELO Ranking System

- Динамический ELO
- Rank progression
- Winrate tracking
- Peak ELO
- MVP statistics
- Match history
- Seasonal rankings

Placement Calibration

- Placement matchmaking
 - Calibration system
 - Automatic placement completion
 - Placement protection
 - Placement-only state
-

Tournament Engine

Поддерживаемые размеры турниров

Players	Status
5	✓
6	✓
7	✓
8	✓
9-15	✓

Players	Status
16	✓

Tournament Features

- Automatic bracket generation
- Qualification rounds
- Quarterfinals
- Semifinals
- Grand Final
- Round of 16
- Automatic BYE logic
- Automatic winner advancement
- Tournament completion system
- Champion rewards

Live Match Veto System

Реализована полноценная live veto система:

- Поочерёдный бан карт
- Автоматическое переключение хода
- Auto-refresh без F5
- Автоматический выбор финальной карты
- Veto validation
- Pending match synchronization

Seasons & Hall of Fame

- Seasonal leaderboard
- Seasonal champions
- Hall of Fame
- Historical snapshots
- Achievement rewards
- Season rewards
- Champion preservation system

Архитектура Проекта

Frontend (React + Vite)

↓

REST API

↓

Backend (FastAPI)



SQLAlchemy ORM



PostgreSQL (Neon)

Tech Stack

Frontend

- React 19
- Vite
- TailwindCSS
- React Router DOM

Backend

- FastAPI
- SQLAlchemy
- Alembic
- Pydantic
- JWT
- Uvicorn

Database

- PostgreSQL (Neon)

Deploy Infrastructure

- Vercel (Frontend)
- Render (Backend)
- Neon PostgreSQL

Структура Проекта

```
CS_League_16-55/  
|  
├── backend/  
│   ├── app/  
│   │   ├── models/  
│   │   ├── routes/  
│   │   ├── schemas/  
│   │   └── utils/
```

```
| | | | | auth.py
| | | | | database.py
| | | | | main.py
| | | | |
| | | | | alembic/
| | | | | requirements.txt
| | | | | runtime.txt
| | | | |
| | | | | frontend/
| | | | | | src/
| | | | | | | components/
| | | | | | | pages/
| | | | | | | hooks/
| | | | | | | layouts/
| | | | | | | utils/
| | | | | |
| | | | | package.json
| | | | | vite.config.js
| | | | |
| | | | | README.md
```

Установка Локально

Clone Repository

```
git clone https://github.com/TursunovJavoxir/CS_League_16-55.git
cd CS_League_16-55
```

Backend Setup

```
cd backend
```

Создание venv

```
python -m venv .venv
```

Активация

Windows

```
.env\Scripts\activate
```

Linux/macOS

```
source .env/bin/activate
```

Install Dependencies

```
pip install -r requirements.txt
```

Backend Environment Variables

Создать `.env`

```
DATABASE_URL=your_postgresql_url  
  
JWT_SECRET_KEY=your_secret_key  
  
JWT_ALGORITHM=HS256  
  
FRONTEND_URL=http://localhost:5173
```

Run Backend

```
uvicorn app.main:app --reload
```



Frontend Setup

```
cd frontend
```

Install Dependencies

```
npm install
```

Frontend Environment Variables

Создать `.env`

```
VITE_API_URL=http://127.0.0.1:8000
```

Run Frontend

```
npm run dev
```

Security Features

- JWT authorization
 - Password hashing
 - Admin protection
 - Duplicate prevention
 - Match validation
 - Placement spam protection
 - Tournament validation
 - Environment variable isolation
-

API Overview

Auth

```
POST /auth/register  
POST /auth/login
```

Players

```
GET    /players
GET    /players/{id}
GET    /players/{id}/matches

POST   /players
POST   /players/bulk
POST   /players/reset-stats
```

Matches

```
POST /matches
POST /matches/{id}/finish
POST /matches/{id}/ban-map
```

Tournaments

```
POST /tournaments
POST /tournaments/{id}/generate-bracket
POST /tournaments/{id}/matches/{id}/finish
```

Реализованные Системы

System	Status
JWT Auth	✓
Admin Panel	✓
ELO System	✓
Placement Matches	✓
Tournament Engine	✓
Live Veto	✓
Hall of Fame	✓
Seasonal Rewards	✓
Achievements	✓

System	Status
Match History	✓
Swagger API	✓
Production Deploy	✓



Roadmap

Planned Features

- WebSocket realtime system
- Team system
- Match analytics
- Spectator mode
- Public API
- Match demo uploads
- Advanced admin dashboard
- CI/CD improvements
- Docker support
- Tournament statistics



Автор

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GitHub:

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License

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